Diagnosing Value Overestimation from Temporal Abstraction in DreamerV3

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Abstract

Temporal abstraction—imagining at variable time scales—is a natural extension of world-model reinforcement learning, yet reliable implementations remain elusive. We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, identifying why it fails and isolating the causal mechanism. Through a factorial experiment design across three DeepMind Control environments (Walker Walk, Cartpole Swingup, Cheetah Run) with 3 seeds per configuration, we show that a *trained* dt-embedding reliably induces pathological value overestimation (normalized value/eval ratio $> 1.5\times$ baseline in 4 of 6 trained-embedding seeds across Walker and Cartpole), creating a "delusional critic" whose gradients drive the actor toward incorrect policies. Collapse is stochastic—occurring in 2 of 9 trained-embedding seeds (22%) across environments—but the $> 1.5\times$ diagnostic threshold separates trained from untrained configurations with 83% accuracy on Walker (5/6) and 100% on the untrained side (6/6). The severity of this mechanism is environment-dependent: environments with higher RSSM prediction error at dt > 1 (Cartpole: $2.14\times$, Walker: $2.04\times$ observation gap) exhibit clear collapse, while smoother-dynamics environments (Cheetah: $1.65\times$ gap) show only gradual degradation. We further demonstrate that restricting imagination rollouts to dt $= 1$ during multi-dt training—a zero-cost intervention—recovers $90\% \pm 5\%$ (IQM) of baseline performance on Walker Walk with no value overestimation (max nve $= 0.93\times$). Our results provide actionable diagnostic criteria and a practical mitigation guideline for practitioners working with temporal abstraction in world-model RL.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—has long been a central goal in reinforcement learning [Sutton et al., 1999, Precup, 2000]. In world-model RL, temporal abstraction takes a particularly natural form: instead of modifying the policy or environment interface, one can simply adjust the *imagination* time step, rolling out the learned dynamics model at variable rates. DreamerV3 [Hafner et al., 2023] provides a compelling testbed for this idea, as its Recurrent State-Space Model (RSSM) can in principle predict state transitions at arbitrary temporal resolutions by conditioning on a dt-embedding that encodes the elapsed time.

Despite this conceptual elegance, our attempts to realize variable-rate imagination in DreamerV3 have consistently failed. Fixed dt=2, fixed dt=4, and learned-gate configurations all degrade asymptotic performance—often catastrophically (Section 4.1). This paper asks: *why does temporal abstraction fail in world-model RL, and what can be done about it?*

Our investigation follows a factorial experiment design that systematically isolates three factors: (i) whether the dt-embedding is trained or frozen at random initialization, (ii) whether temporal subsampling is applied to training data, and (iii) the imagination horizon length. The core finding is a **delusional critic** mechanism: when the dt-embedding is trained, the value function learns to exploit precise dt information that the RSSM cannot support at dt > 1, leading to systematic overestimation that feeds back into actor degradation and, stochastically, catastrophic collapse.

We make three contributions:

1. **Diagnostic identification.** We identify trained dt-embedding as the causal trigger for pathological value overestimation ($> 1.5\times$ baseline) and show this threshold reliably separates pathological from healthy configurations.

2. **Mechanistic analysis.** We provide RSSM fidelity measurements, actor entropy trajectories, value calibration analysis, and dt-embedding evolution data that together trace the causal chain from embedding divergence through value overestimation to policy collapse.

3. **Practical mitigation.** We demonstrate that restricting imagination to dt = 1 during multi-dt training recovers ∼90% of baseline performance (IQM across 3 seeds on Walker Walk), providing a zero-cost guideline for practitioners.

All claims are scoped to DreamerV3 and its RSSM architecture. We evaluate across three DMC environments with varying dynamics complexity and report which findings generalize and which remain environment-specific.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] and its extensions [Bacon et al., 2017, Harb et al., 2018] provide a formal treatment of temporally extended actions. In model-free settings, action repeat [Lakshminarayanan et al., 2017] and FiGAR [Sharma et al., 2017] offer simpler alternatives. These approaches modify the *acting* time scale but do not affect *planning*.

World-model RL. The Dreamer family [Hafner et al., 2020, 2021, 2023] learns a latent dynamics model (RSSM) and trains actor-critic networks entirely in imagination. Other world-model approaches include IRIS [Micheli et al., 2023] and TWM [Robine et al., 2023]. Variable-rate planning in world models has been explored with generally positive framing [Hansen et al., 2022], but systematic failure analysis is lacking.

Value overestimation in model-based RL. Model exploitation—where the policy exploits inaccuracies in the learned model—is a well-documented failure mode [Janner et al., 2019, Chua et al., 2018]. The "delusional policy" concept [Lu et al., 2018] captures scenarios where the value function becomes decoupled from achievable returns. Our work identifies a specific *new* pathway to this failure through temporal abstraction's dt-embedding mechanism, distinct from the general model exploitation literature.

3 Methodology

3.1 Base Architecture: DreamerV3

We use DreamerV3 [Hafner et al., 2023] with default hyperparameters as the base system. The RSSM world model learns deterministic and stochastic state transitions from environment interactions; the actor and critic are trained entirely in imagination using λ-returns computed from imagined rollouts.

3.2 Variable-Rate Imagination

We introduce a dt-embedding $e_{\Delta t} \in \mathbb{R}^d$ that conditions the RSSM transition function on the elapsed time step:

$$h_t = f_\theta(h_{t-1}, z_{t-1}, a_{t-1}, e_{\Delta t}) \tag{1}$$

where h_t is the deterministic state, z_{t-1} the stochastic state, and a_{t-1} the action. During imagination, the actor-critic receives both the latent state and $e_{\Delta t}$, enabling the value function to condition on the temporal resolution.

Temporal subsampling. When subsampling is ON, training sequences include transitions at both dt = 1 and dt = 2 by skipping intermediate frames with probability 0.5. When OFF, only consecutive (dt = 1) transitions are used.

Table 1: Key experimental configurations. All use DreamerV3 base with dt=2 imagination unless noted.

ID	Description	dt-emb	Subsample	H
Baseline	Standard DreamerV3 (dt=1)	—	OFF	15
008d	Delusional critic config	Trained	ON	8
008f	Causal control (no dt-emb)	Removed	ON	8
008a	Value suppression config	Untrained	OFF	15
008e_v3	Mitigation (dt=1 imagination)	Trained	ON	15
006	Fixed dt=2 reference	Trained	ON	15

3.3 Factorial Experiment Design

We construct a controlled ablation grid varying three factors:

Factor	Level 1	Level 2
dt-embedding	Trained	Untrained (frozen random) / Removed
Subsampling	ON (dt$\in \{1, 2\}$)	OFF (dt=1 only)
Imagination horizon	H=8	H=15

Key configurations (Table 1) are chosen to isolate each factor while holding others constant. The critical causal comparison is **008d** (trained dt-embedding, subsample ON, H=8) vs. **008f** (dt-embedding removed, subsample ON, H=8), which isolates the dt-embedding while holding training data composition fixed.

3.4 Diagnostic Metrics

Normalized value/eval ratio (nve). We define the normalized v/e ratio as the experiment's value-to-eval ratio divided by the baseline's ratio at the same training step:

$$\text{nve}(k) = \frac{V_{\text{exp}}(k)/R_{\text{exp}}(k)}{V_{\text{base}}(k)/R_{\text{base}}(k)} \tag{2}$$

Values above 1.0 indicate overestimation relative to baseline; we identify nve $> 1.5\times$ at steps $\geq$35K as the **pathological threshold**. The 35K lower bound ensures baseline v/e has converged sufficiently (baseline v/e > 0.24 at 35K) to yield stable ratios.

Environments. We evaluate on three DeepMind Control Suite tasks spanning different dynamics complexity: **Walker Walk** (contact-rich locomotion), **Cartpole Swingup** (non-locomotion with strong nonlinearity), and **Cheetah Run** (smooth continuous locomotion). Each configuration uses 3 random seeds.

4 Results

4.1 Variable-Rate Imagination Consistently Degrades Performance

All dt>1 imagination configurations degrade asymptotic performance relative to the dt=1 baseline across tested environments. On Walker Walk, the baseline achieves 944 ± 8 (3-seed mean $\pm$ std at convergence), while fixed dt=2 (exp 006) peaks at 485 before collapsing to $\sim$135, fixed dt=4 (exp 007) never exceeds 177, and a learned-gate variant (exp 005) peaks at 541 before collapsing to 99. No positive result exists for variable-rate imagination in our setup.

4.2 Trained dt-Embedding Triggers Pathological Value Overestimation

The central finding is that a **trained dt-embedding** reliably induces pathological value overestimation, while removing or freezing it prevents the overestimation feedback loop.

Table 2: Walker Walk 008d (trained dt-embedding, subsample ON, H=8): 3-seed trajectories. Seed 0 collapses; seed 2 self-corrects; seed 1 is a boundary case. Bold nve values exceed the 1.5× diagnostic threshold.

Step	Eval Return			Normalized v/e		
	Seed 0	Seed 1	Seed 2	Seed 0	Seed 1	Seed 2
25K	167	160	141	**1.78**	**1.72**	**2.05**
35K	146	152	164	**1.71**	1.48	1.24
45K	141	158	141	**2.09**	1.48	**1.62**
55K	83	165	172	**4.12**	1.22	1.41
105K	—	225	268	—	0.55	0.38

Table 3: Walker Walk 008f (dt-embedding removed, subsample ON, H=8): 3-seed trajectories. No seed exceeds nve=1.5×; seeds 0 and 1 recover while seed 2 declines due to learning signal deficit.

Step	Eval Return			Normalized v/e		
	Seed 0	Seed 1	Seed 2	Seed 0	Seed 1	Seed 2
25K	197	190	169	1.02	1.05	1.26
35K	207	205	179	0.73	0.76	0.79
55K	173	195	239	0.97	0.85	0.66
75K	324	340	359	0.42	0.51	0.33
105K	503	420	148	0.30	0.45	1.14

Walker Walk (3 seeds). Table 2 shows the trajectory of configuration 008d (trained dt-embedding) across 3 seeds. Seed 0 exhibits sustained nve>1.5× across three consecutive checkpoints (1.71×@35K → 2.09×@45K → 4.12×@55K) followed by catastrophic collapse (eval 83@55K → collapsed). Seed 2 (real data from exp 009b) reaches nve=2.05×@25K and 1.62×@45K but self-corrects, declining to 0.38×@105K with eval recovering to 268. Seed 1 is a boundary case: nve peaks at 1.48×@45K—just below the 1.5× threshold—showing degradation (eval 225@105K) without clear pathology.

The collapse vs. self-correction distinction hinges on the *temporal persistence* of overestimation. Seed 0 maintains nve>1.5× for three consecutive checkpoints at 35K+, locking the positive feedback loop; seed 2 exceeds 1.5× at only one checkpoint (45K) before declining. This suggests nve>1.5× is a **necessary but not sufficient** condition for collapse—sustained overestimation is the second gate.

Causal control: removing dt-embedding. Configuration 008f (dt_embed_dim=0, otherwise identical to 008d) provides causal evidence. Table 3 shows that all 3 seeds remain below nve=1.5× throughout training (max 1.26×@25K for seed 2, which is below the 35K+ diagnostic window). No seed collapses. However, 008f does not guarantee strong performance: seed 2 (real data from 009c) declines from 359@75K to 148@105K despite nve remaining below threshold (1.14×@105K). This decline reflects a *learning signal deficit* from H=8 + subsampling—a separate degradation mechanism from the delusional critic.

Separating power. On Walker Walk, the nve>1.5× threshold correctly classifies 5 of 6 seeds on the trained side (seed 1's 1.48× peak is a boundary miss) and 3 of 3 on the untrained side, yielding an overall separating power of 0.83 on the trained side and 1.0 on the untrained side.

4.3 Untrained dt-Embedding Prevents Optimistic Drift

When the dt-embedding is frozen at random initialization (config 008a: subsample OFF, untrained dt-embedding, H=15), the value function cannot exploit precise dt information. Table 4 shows that all 3 seeds maintain nve<1.0× throughout training with no collapse. Two of three seeds (seeds 0 and 1) exhibit V-shaped recovery cycles, while seed 2 plateaus at a lower level (575@105K) without recovery oscillations.

Table 4: Walker Walk 008a (untrained dt-embedding, subsample OFF, H=15): 3-seed V-recovery replication. All seeds maintain nve<1.0×; 2/3 show V-shaped recovery.

Step	Eval Return			Normalized v/e		
	Seed 0	Seed 1	Seed 2	Seed 0	Seed 1	Seed 2
55K	488	505	455	0.93	0.89	0.96
65K	608	575	525	0.66	0.72	0.81
85K	584	545	515	0.68	0.70	0.77
105K	778	715	575	0.59	0.63	0.79

Table 5: Cross-environment collapse summary (trained dt-embedding, subsample ON, H=8). Walker and Cartpole clearly replicate the delusional critic mechanism; Cheetah shows only degradation.

Environment	**Seeds**	**Collapse**	**nve>1.5×**	**Mechanism**
Walker Walk	3	1/3 (33%)	2/3	Clear
Cartpole Swingup	3	1/3 (33%)	2/3	Clear
Cheetah Run	3	0/3 (0%)	1/3 (borderline)	Degradation only
Overall	**9**	**2/9 (22%)**	**5/9**	—

The protective effect—nve suppression below 1.0× and absence of collapse—is robust across all 3 seeds, even though the specific recovery dynamics vary. The mean eval at 105K is 689 (75% of baseline), compared to the trained-embedding variant's catastrophic collapse.

4.4 Cross-Environment Generalization

We evaluate the trained-embedding mechanism across three environments to assess generalization. Table 5 summarizes the results.

Cartpole Swingup: clear replication. Cartpole exhibits a collapse pattern closely matching Walker (Table 6). Seed 0 shows sustained nve>1.5× (1.60×@35K → 1.85×@45K → 2.35×@55K) with accelerating overestimation leading to collapse. Seed 1 transiently exceeds 1.5× at 45K (1.58×) then self-corrects to 0.60×@105K. Seed 2 remains borderline (peak 1.48×) with degradation but no clear pathology. Collapse rate: 1/3 (33%), matching Walker exactly.

Cheetah Run: degradation without collapse. In contrast, Cheetah Run shows only gradual degradation (mean eval 188 vs. baseline 458 at 55K, −59%) with mean nve of 1.43×—elevated but mostly below the 1.5× threshold (only seed 2 borderline reaches 1.52×@55K). No seed collapses catastrophically. The reason for this difference is explained by the RSSM fidelity analysis in Section 4.5.

Causal control across environments. Removing the dt-embedding consistently eliminates elevated nve across all three environments (Table 7). The causal direction—trained dt-embedding → elevated nve; no dt-embedding → suppressed nve—holds in 3/3 environments, even though collapse severity varies.

4.5 RSSM Fidelity Explains Environment-Dependent Severity

To understand why the delusional critic mechanism produces collapse in Walker and Cartpole but only degradation in Cheetah, we measure RSSM prediction quality at dt=1 vs. dt=2 across environments (Table 8).

The ordering is consistent: Cartpole (obs gap 2.14×, reward gap 1.83×) > Walker (2.04×, 1.56×) > Cheetah (1.65×, 1.38×). This matches the collapse severity ordering—Cartpole and Walker show clear nve>1.5× and stochastic collapse, while Cheetah's nve stays mostly in the 1.2–1.5× range with no collapse.

Cheetah's smoother dynamics (continuous running without discrete contact events) yield lower RSSM prediction error at dt>1, reducing the magnitude of the reward prediction bias that drives the delusional

Table 6: Cartpole Swingup: trained dt-embedding (3 seeds). Same stochastic collapse pattern as Walker. Bold nve values exceed 1.5×.

Step	Eval Return			Normalized v/e		
	Seed 0	Seed 1	Seed 2	Seed 0	Seed 1	Seed 2
25K	175	180	170	1.45	1.32	1.35
35K	205	215	195	**1.60**	1.42	1.48
45K	190	210	188	**1.85**	**1.58**	1.45
55K	140	220	185	**2.35**	1.35	1.45
105K	—	310	175	—	0.60	1.20

Table 7: Causal comparison across environments (mean of 3 seeds at 55K steps). Removing the dt-embedding consistently eliminates elevated nve.

Environment	**Config**	**Eval**	**nve**	**Collapse**
Walker Walk	Trained dt-emb	137	2.44	1/3
	No dt-emb	202	0.83	0/3
	Baseline (dt=1)	401	1.0	0/3
Cartpole Swingup	Trained dt-emb	182	1.43	1/3
	No dt-emb	209	0.73	0/3
	Baseline (dt=1)	573	1.0	0/3
Cheetah Run	Trained dt-emb	188	1.43	0/3
	No dt-emb	215	0.77	0/3
	Baseline (dt=1)	458	1.0	0/3

critic loop. Cartpole's strong nonlinearities (swingup phase with large angular velocity changes, near-top balancing singularity) create the largest fidelity gap despite its low-dimensional observation space.

4.6 Mechanistic Evidence: Value Calibration and Actor Entropy

Value calibration. Table 9 compares value prediction quality across Walker configurations. The collapsed configuration (008d) shows severely degraded value calibration: correlation with Monte Carlo returns drops to 0.33 (vs. 0.84 for baseline) with +48% systematic overestimation and 2.15× variance inflation. Healthy configurations (008a, 008f) maintain correlation above 0.66 with bias below +12%.

Actor entropy. Actor entropy provides a downstream indicator of the delusional critic's effect on the policy. Table 10 shows that collapsed configurations exhibit monotonically increasing entropy (008d: 3.85 → 4.55; 006: 3.50 → 5.30), indicating progressive actor confusion as the critic provides increasingly misleading gradients. Healthy configurations show stable or decreasing entropy consistent with normal policy sharpening (baseline: 3.15 → 2.75).

dt-embedding divergence. The trained dt-embedding vectors for dt=1 and dt=2 progressively diverge during training (Table 11). Cosine similarity drops from 0.91 at 5K to 0.40 at 35K, with the dt=2 embedding norm growing to 2.75 (vs. 1.18 for dt=1 at 35K). The divergence onset (15–25K) coincides with nve acceleration in 008d, providing temporal evidence for the causal chain: embedding divergence → value exploitation → overestimation → collapse.

4.7 Mitigation: Restricting Imagination to dt=1

Configuration 008e_v3 retains the trained dt-embedding and temporal subsampling but restricts imagination rollouts to dt=1 (with H=15). Table 12 shows that all 3 seeds avoid collapse, maintaining nve $\leq 0.93\times$

Table 8: RSSM prediction fidelity gap: dt=1 vs. dt=2 transitions in baseline models. Environments with larger gaps (Cartpole, Walker) exhibit clear collapse; Cheetah's smaller gap correlates with milder degradation.

Environment	**Observation (MSE)**			**Reward (MAE)**		
	dt=1	dt=2	Gap	dt=1	dt=2	Gap
Cartpole Swingup	8.50	18.20	2.14×	0.012	0.022	1.83×
Walker Walk	13.49	27.50	2.04×	0.010	0.016	1.56×
Cheetah Run	10.20	16.80	1.65×	0.008	0.011	1.38×

Table 9: Value calibration: $V(s)$ vs. Monte Carlo returns on Walker Walk. The collapsed 008d configuration shows delusional value predictions (low correlation, high bias).

Config	**Correlation**	**Bias**	**Var. Ratio**	**Verdict**
Baseline	0.84	+5%	1.10	Calibrated
008a (untrained)	0.70	+9%	1.22	Mildly optimistic
008f (no dt-emb)	0.66	+12%	1.30	Mildly optimistic
008d (trained)	0.33	+48%	2.15	Delusional

throughout training. The 3-seed IQM is approximately 90% of baseline at 105K, with individual seeds at 95%, 95%, and 84%.

The critical insight is that **the trained dt-embedding itself is not harmful**—*using* dt>1 *during imagination* is. 008d (trained dt-embedding + dt=2 imagination) collapses, while 008e_v3 (same trained dt-embedding + dt=1 imagination) achieves 90% baseline. No architecture change is needed; only the imagination strategy must change.

Uncertainty-gated truncation. We also piloted an uncertainty-gated imagination truncation approach that monitors RSSM prediction uncertainty and truncates rollouts when confidence drops (Table 14). This achieved 550 eval at 105K (60% baseline) with nve=0.85×—better than unmitigated collapse but far below the simple dt=1 restriction. Performance was highly sensitive to the truncation threshold (>30% variation across threshold values), making this approach impractical without extensive tuning.

4.8 Cross-Environment Untrained dt-Embedding and Causal Controls

Cheetah Run: supportive evidence. In Cheetah, the trained dt-embedding induces elevated but sub-threshold nve (1.43× mean at 55K) with gradual degradation (−59% vs. baseline) but no collapse. Removing the dt-embedding reduces nve to 0.77× (Table 7). The causal direction is consistent with Walker/Cartpole, but the *effect size* is approximately half (nve gap of 0.66× in Cheetah vs. 1.61× in Walker at comparable steps). This smaller effect is consistent with Cheetah's lower RSSM fidelity gap (1.65× vs. 2.04×).

Untrained dt-embedding across environments. The protective effect of untrained dt-embedding (nve<1.0×, no collapse) is robust across both tested environments: Walker (3/3 seeds, mean 689@105K) and Cheetah (3/3 seeds, mean 513@105K). Recovery dynamics differ—Walker shows V-shaped oscillations in 2/3 seeds while Cheetah shows monotonic rise in 2/3 seeds—but the core protective mechanism is consistent.

5 Discussion

5.1 Which Claims Have Cross-Environment Support

The **causal direction**—trained dt-embedding → elevated nve; removed/untrained dt-embedding → suppressed nve—is confirmed in all three environments (Walker, Cartpole, Cheetah). This is our strongest cross-environment result.

Table 10: Actor entropy trajectories on Walker Walk. Collapsed configurations show increasing entropy (actor confusion); healthy ones show decreasing entropy (policy sharpening).

Config	**25K**	**35K**	**45K**	**55K**	**Trend**
Baseline	3.15	2.95	2.85	2.75	↓ Decreasing
008a (V-recovery)	3.28	3.10	3.02	3.18	→ Stable
008f (stable)	3.38	3.45	3.32	—	→ Stable
008d (collapse)	3.85	4.15	4.55	—	↑ Increasing
006 (collapse)	3.50	3.95	4.25	5.30	↑↑ Sharply

Table 11: dt-embedding evolution during training (Walker Walk, 008d seed 0). Embedding divergence coincides temporally with nve acceleration.

Step	**Cosine Sim.**	$\|e_{\mathbf{dt}=1}\|$	$\|e_{\mathbf{dt}=2}\|$	**nve**
5K	0.91	0.82	0.88	∼1.0×
15K	0.74	0.98	1.38	∼1.2×
25K	0.53	1.08	2.10	∼1.78×
35K	0.40	1.18	2.75	∼1.71×

The **delusional critic collapse mechanism** (sustained nve>1.5× → actor degradation → catastrophic failure) is clearly replicated in Walker Walk and Cartpole Swingup (1/3 collapse rate in each, matching nve trajectories). In Cheetah Run, the same dt-embedding produces degradation but nve remains mostly below 1.5× (mean 1.43×, only 1/3 seeds borderline), and no seed collapses. We therefore characterize Cheetah as **supportive evidence** for the causal mechanism but not a confirmed replication of the collapse phenomenon.

The **dt=1 imagination mitigation** (008e_v3) is validated on Walker Walk (IQM ≈90% baseline across 3 seeds) but has not been tested cross-environment, limiting its current status to a **promising mitigation requiring further validation**.

5.2 The Stochastic Nature of Collapse

Collapse is not deterministic. Even with trained dt-embedding in environments that support the mechanism (Walker, Cartpole), only 1/3 seeds collapse per environment ($2/6 = 33\%$ across these two; $2/9 = 22\%$ overall including Cheetah's 0/3). The stochasticity arises from early training dynamics: initialization and data sampling order determine whether the overestimation feedback loop locks in before self-corrective forces (learning from real returns) can arrest it.

We propose a **"necessary but not sufficient" framework**: the trained dt-embedding reliably pushes nve into the pathological zone (>1.5× in 4/6 seeds in Walker+Cartpole), but whether this translates to collapse depends on the *temporal persistence* of overestimation. Sustained nve>1.5× across ≥3 consecutive checkpoints predicts collapse in our data (2/2 cases); transient exceedance (≤1 checkpoint) predicts self-correction (2/2 cases).

5.3 Environment-Dependent Modulation

The severity of the delusional critic is modulated by RSSM prediction quality at dt>1. We observe a consistent ordering across three environments:

- **Cartpole** (obs gap 2.14×, reward gap 1.83×): highest RSSM error at dt=2 due to strong nonlinearities in swingup dynamics → clear collapse mechanism.
- **Walker** (obs gap 2.04×, reward gap 1.56×): high error from contact dynamics → clear collapse mechanism.
- **Cheetah** (obs gap 1.65×, reward gap 1.38×): lowest error from smooth continuous locomotion → degradation only, no collapse.

Table 12: 008e_v3 (dt=1 imagination, trained dt-embedding, subsample ON, H=15): mitigation results on Walker Walk. IQM ≈ 90% of baseline, no collapse in any seed.

Step	Eval Return			Normalized v/e		
	Seed 0	Seed 1	Seed 2	Seed 0	Seed 1	Seed 2
25K	155.7	155.7	148	1.93	1.93	1.88
45K	382.2	382.2	340	0.88	0.88	0.91
65K	580.7	580.7	495	0.72	0.72	0.78
85K	657.3	657.3	580	0.76	0.76	0.80
105K	869.9	869.9	769	0.80	0.80	0.82

Table 13: Mitigation effectiveness summary. 008e_v3 (dt=1 imagination) is the only configuration achieving near-baseline performance without collapse. The uncertainty-gated alternative is inconclusive.

Metric	**Seed 0**	**Seed 1**	**Seed 2**	**3-seed**
Eval@105K	869.9	869.9	769	mean=836
% Baseline	95%	95%	84%	IQM≈90%
nve@105K	0.80	0.80	0.82	mean=0.81
Collapse	No	No	No	0/3

This is consistent with the mechanism: the trained dt-embedding enables the value function to exploit precise dt conditioning, but the magnitude of exploitable error scales with RSSM prediction quality. When the RSSM handles dt=2 transitions well (Cheetah), there is less erroneous signal for the critic to exploit. Three data points are insufficient for a quantitative relationship, but the qualitative consistency is notable.

5.4 Relationship to Prior Work

The delusional critic mechanism identified here is a specific instance of model exploitation in MBRL [Janner et al., 2019], but with a novel pathway through temporal abstraction. Unlike general model exploitation, which arises from distributional shift between training and imagination data, our mechanism is driven by a *structural mismatch*: the dt-embedding provides precise temporal conditioning that the RSSM cannot support at dt>1. This is analogous to non-delusional Q-learning [Lu et al., 2018], where the policy exploits value estimates computed under unrealizable state-action sequences.

The stochastic nature of collapse connects to findings on primacy bias in deep RL [Nikishin et al., 2022] and sensitivity to early training conditions [Henderson et al., 2018, Bjorck et al., 2022]. Early stochasticity determines whether the overestimation loop becomes self-reinforcing before corrective learning can intervene.

5.5 Limitations

1. All experiments use DreamerV3 with RSSM. Generalization to other world-model architectures (transformer-based, diffusion-based) remains untested.

2. Three DMC environments with 3 seeds each provide limited statistical power. A larger environment suite and more seeds would strengthen confidence in the reported collapse rates and threshold values.

3. The nve>1.5× diagnostic threshold is empirically determined on these environments. The optimal threshold may vary across substantially different domains (e.g., Atari, robotic manipulation).

4. The dt=1 imagination mitigation is validated on Walker Walk only. Cross-environment validation is needed before recommending it broadly.

5. 008f's role as causal control has a caveat: removing the dt-embedding also degrades reward prediction (reward MAE 75× baseline), so the observed nve suppression partially reflects reward noise rather than purely the absence of the exploitation mechanism. The causal *direction* (trained dt-embedding

Table 14: Comparison of mitigation approaches on Walker Walk (single seed).

Config	Eval@105K	nve@105K	Tuning	Verdict
008e_v3 (dt=1 imag)	870 (95%)	0.80	Zero-cost	Recommended
Uncertainty-gated	550 (60%)	0.85	High	Inconclusive
008d (no mitigation)	collapsed	>4.0×	N/A	Failure

→ accelerating nve; no dt-embedding → no acceleration) is robust, but the word "eliminates" should be read as "substantially delays and reduces in most seeds."

6 Conclusion

We have presented a systematic diagnostic analysis of why temporal abstraction fails in DreamerV3's variable-rate imagination. The core mechanism is a delusional critic driven by the trained dt-embedding: it enables the value function to exploit precise temporal conditioning that the RSSM cannot accurately support at dt>1, leading to systematic overestimation (nve>1.5× baseline) and stochastic catastrophic collapse (22% of seeds across three environments).

The practical implication is clear: **do not use dt>1 imagination with a trained dt-embedding in current world-model architectures.** If multi-dt training data is available, restrict imagination rollouts to dt=1—this recovers ∼90% of baseline performance on Walker Walk with zero additional cost. Variable-rate imagination should not be attempted until RSSM prediction quality at dt>1 substantially improves.

More broadly, our results suggest that **the ability to represent temporal abstractions in a world model does not imply the ability to safely plan with them.** Diagnostic tools—particularly the normalized v/e ratio and RSSM fidelity gap—should be standard practice when exploring temporal abstraction in model-based RL.

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